Figures that preserve evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Teach the evidence-to-conclusion chain for Figures that preserve evidence

Preserve units, case counts and uncertainty limits without making a figure stand in for missing evidence.

Keep one habit: Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of figures that preserve **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of negative results and claim revision

Reference example, not the only acceptable wording: Teacher explanation: Negative results report that the **evidence** did not meet a declared expectation.

They can reveal design limits or useful constraints without proving no effect anywhere. Compare results with the original criterion and uncertainty assumptions.

Preserve the failed hypothesis, discuss plausible **mechanism**s and label revised questions as new work.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of inspecting failures and anomalies

Reference example, not the only acceptable wording: Teacher explanation: Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of power and resource tradeoffs

Reference example, not the only acceptable wording: Teacher explanation: Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Pictured end goal

Explain figures that preserve **evidence** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Negative results and claim revision

Inspecting failures and anomalies

Power and resource tradeoffs

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of **evidence**, not extra **evidence**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of figures that preserve **evidence**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

Definitions and complete contract

Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of **evidence**, not extra **evidence**.

Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

Figure item	Complete supplied exercise contract
Main rates	A 80%, B 82%; raw case counts are not supplied
Main problematic axis	Bars start at 79% without adequate disclosure
Correct main difference	Two percentage points; not a several-fold increase in the actual rates
Separate picture	A six correct, B eight correct on the same ten cases
Picture axis	Correct-count axis starts at zero; denominator ten visible
Uncertainty	No interval or raw study design supplied; do not fabricate error bars

Figure item	Complete supplied exercise contract
Interpretation	A clear figure does not establish sampling quality, causal advantage or population performance

Main evidence ledger and checked key

Main figure calculation	Complete value	Meaning
A rate	80%	Supplied rate; underlying denominator unknown
B rate	82%	Supplied rate; underlying denominator unknown
Absolute difference	82% – 80% = 2 percentage points	Correct rate-scale difference
Relative increase	2/80 = 2.5%	Different from two percentage points
Heights above hidden 79% baseline	A: 1 point; B: 3 points	Threefold drawn heights distort the actual rate ratio
Repair	Zero-baseline percentage bars or clearly labelled difference plot	Retain missing case-count and uncertainty information

Main 80% versus 82% differs by two percentage points, or 2.5% relative to A; raw case counts are unknown. Hidden 79% start gives drawn heights one and three, misleadingly threefold. Picture six/eight of ten gives 60%/80%, difference twenty points. Hidden start at five again produces heights one/three instead of true counts six/eight.

Independent A 8/10 = 80%, B 9/15 = 60%; raw nine versus eight hides different denominators. Plot rates with totals and context. Different case sets/eligibility are not established as matched, so do not claim a controlled superiority from rates alone. No uncertainty or significance can be fabricated.

Figures that preserve evidence — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Rates A 80%, B 82% are plotted on bars starting 79%, making B look several times larger.	Label axes, units, cases and scope. For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.	Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled difference plot.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**. Qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger.

Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and misconception probes

Figure audit	Checked interpretation
Main 80% versus 82%	Two percentage points, not two percent relative
Underlying main denominator missing	Do not invent eighty or eighty-two of one hundred cases
Picture counts six/eight with common ten	Show denominator and zero count baseline
Bar axis truncated without clear disclosure	Exaggerates relative bar lengths
Focused difference plot	Use an explicit difference scale and explain units
Uncertainty not supplied	Do not invent intervals or significance stars
Counts eight versus nine with different totals	Show rates and both denominators
Different case sets	Rate comparison alone does not prove a fair controlled advantage

Separate pictured facts and full reasoning

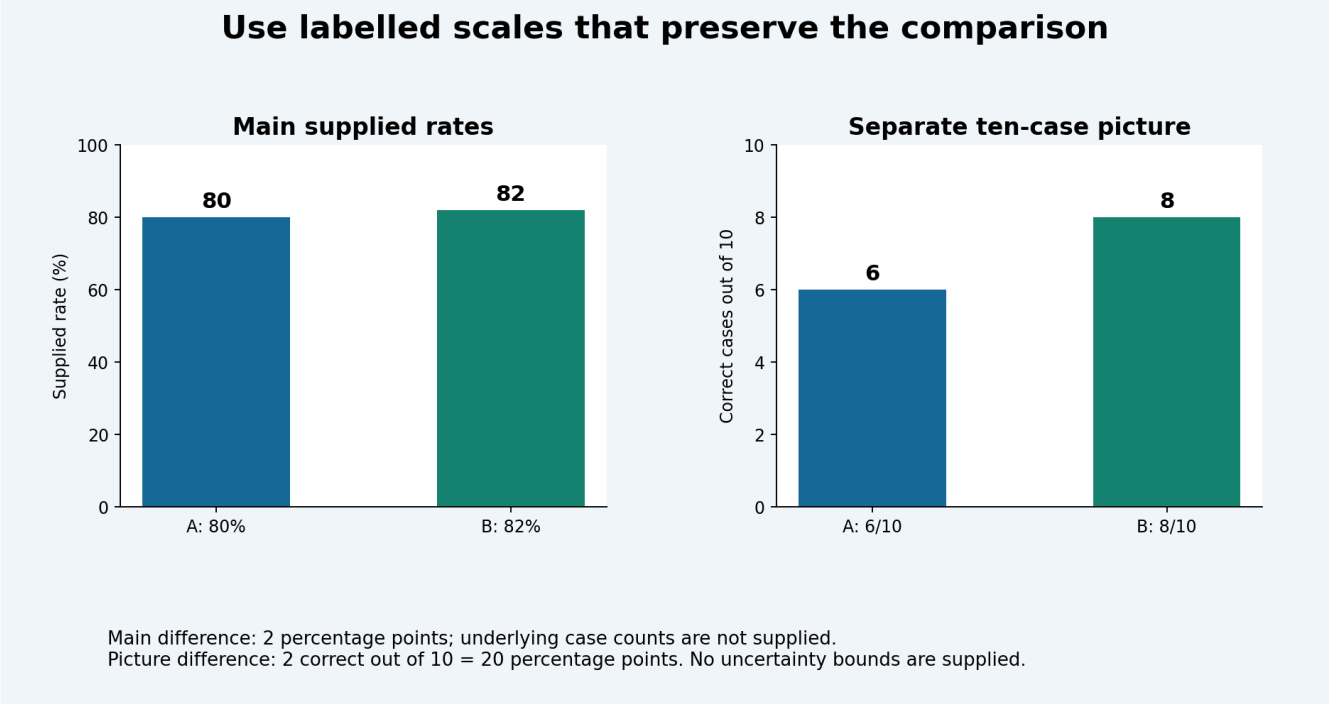
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

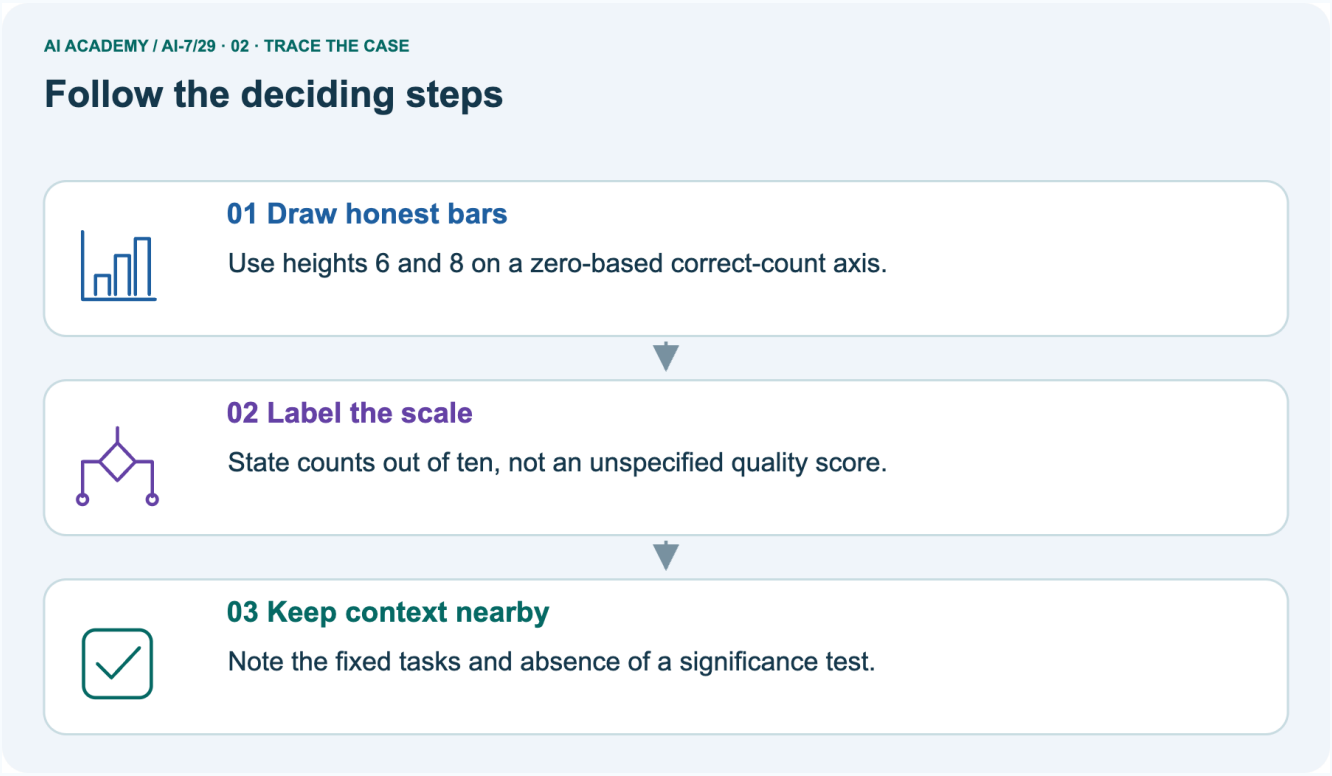
The separate picture compares six versus eight correct on the same ten cases. A is 60%, B 80%; the difference is two correct or twenty percentage points. Use count bars from zero to ten and label the common ten-case denominator. If the bars secretly start at five, their visible heights are one and three, creating a threefold height ratio even though the actual count ratio is eight divided by six.

The new native figure keeps main supplied percentages separate from the ten-case count example. No uncertainty bounds are supplied, and a well-labelled chart does not turn ten cases into a population or causal conclusion.

Separate ten-case picture	A	B
Correct count	6	8
Evaluated cases	10	10
Observed rate	60%	80%
Difference	Two extra correct	Twenty percentage points
Zero-baseline count height	6	8
Hidden start-at-five height	1	3
Why hidden truncation misleads	Drawn heights appear threefold	Actual count ratio 8/6 is not threefold
Uncertainty / population inference	Not supplied	Not supplied



Picture trace and changed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Draw honest bars	Use heights 6 and 8 on a zero-based correct-count axis.
2. Label the scale	State counts out of ten, not an unspecified quality score.
3. Keep context nearby	Note the fixed tasks and absence of a significance test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

A well-labelled chart still cannot establish a causal or population claim from ten cases.

Change: Start the axis at 5 without making the truncation obvious.

Prediction: The bars exaggerate the visual contrast and require correction or clear disclosure.

Reason: Visible heights no longer correspond to the full count scale.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Axis scale	The mapping between values and plotted positions
Denominator label	The sample size underlying a proportion
Difference plot	A display of a defined comparison magnitude

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

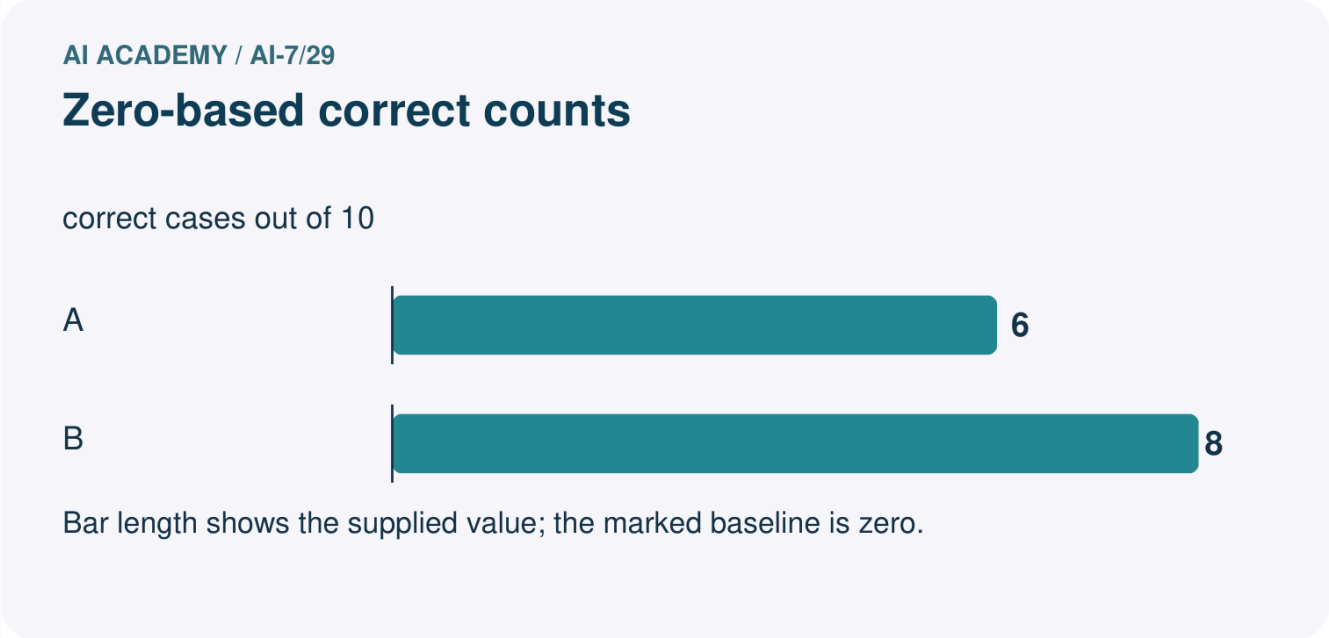
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Figures that preserve evidence

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/29

Figures that preserve evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw honest bars

Use heights 6 and 8 on a zero-based correct-count axis.



2. Label the scale

State counts out of ten, not an unspecified quality score.



3. Keep context nearby

Note the fixed tasks and absence of a significance test.

CHANGE ONE THING

Start the axis at 5 without making the truncation obvious.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Figures that preserve **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

Use heights 6 and 8 on a zero-based correct-count axis. State counts out of ten, not an unspecified quality score. Note the fixed tasks and absence of a significance test.

Changed case: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

Changed-case reasoning: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main 80% versus 82% differs by two percentage points, or 2.5% relative to A; raw case counts are unknown. Hidden 79% start gives drawn heights one and three, misleadingly threefold. Picture six/eight of ten gives 60%/80%, difference twenty points. Hidden start at five again produces heights one/three instead of true counts six/eight.

Independent A 8/10 = 80%, B 9/15 = 60%; raw nine versus eight hides different denominators. Plot rates with totals and context. Different case sets/eligibility are not established as matched, so do not claim a controlled superiority from rates alone. No uncertainty or significance can be fabricated.

Diagnose the current misconceptions

The worked result establishes every future case: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

Teacher explanation: Rates**80%** versus**60%**; count-only plot hides unequal denominators. Show both counts and rates with sample sizes.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Figures should preserve denominators, scales and uncertainty rather than exaggerate the preferred comparison. A plot is a representation of **evidence**, not extra **evidence**. Label axes, units, cases and scope.

For simple comparative bars use a zero baseline or explicitly justify another plot. Show condition breakdowns and unassessed counts where relevant.

W2 Trace the worked case

Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**.

W3 Audit the limit

A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Visual choices must reflect the intended metric and cannot imply causal certainty from association.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**. Qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

Teacher explanation: Rates**80%** versus**60%**; count-only plot hides unequal denominators. Show both counts and rates with sample sizes.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Rates**80%** versus**60%**; count-only plot hides unequal denominators. Show both counts and rates with sample sizes. Require intermediate reasoning and the stated scope.

Fresh retry: Two estimates have intervals but chart omits them and claims certainty. Expected reasoning: Preserve the valid uncertainty intervals/assumptions; absence in a chart does not remove sampling variability.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled figures that preserve **evidence evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of figures that preserve **evidence**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Writing the methods section. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/30, ai-7/32, ai-7/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of figures that preserve **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Difference is 2 percentage points; truncated bars exaggerate magnitude. Use zero-baseline bars or a clearly labelled **difference plot**.

Check every learner before discussion: Use the worked case for Figures that preserve **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Start the axis at 5 without making the truncation obvious. Require a **prediction** and a reason before comparing. Expected result: The bars exaggerate the visual contrast and require correction or clear disclosure.

Reason: Visible heights no longer correspond to the full count scale.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

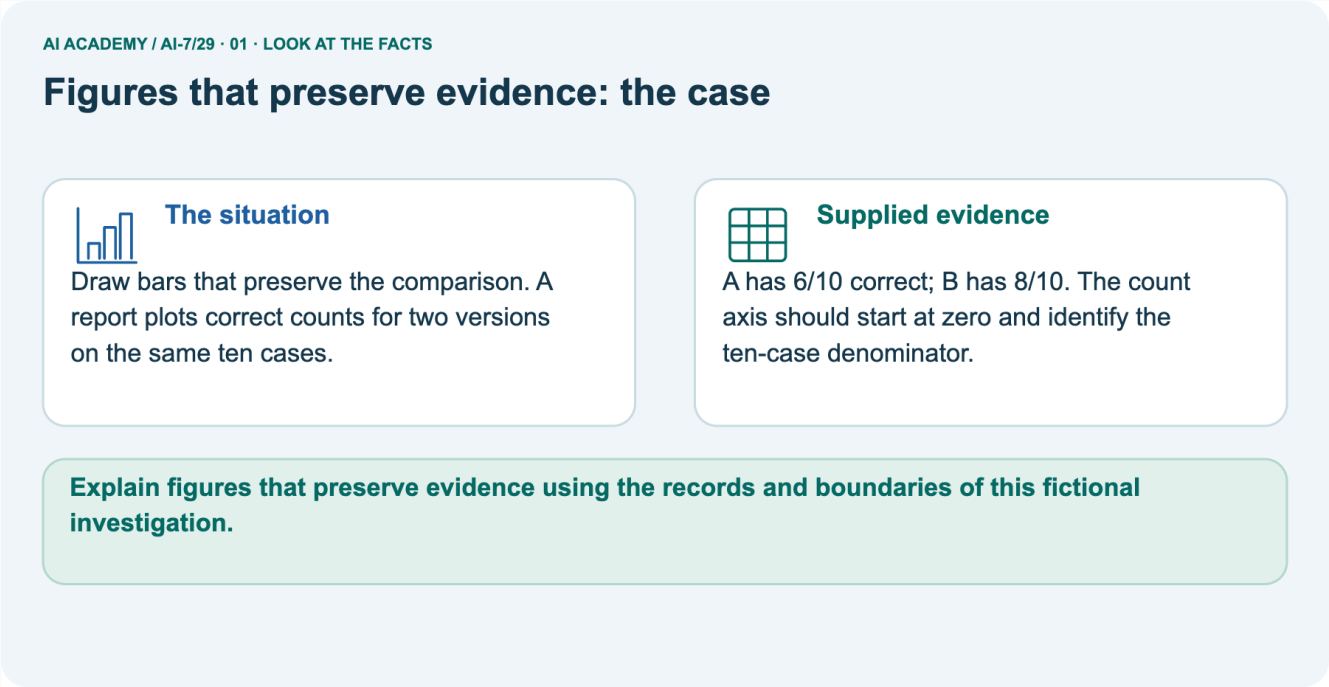
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Figures that preserve evidence: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

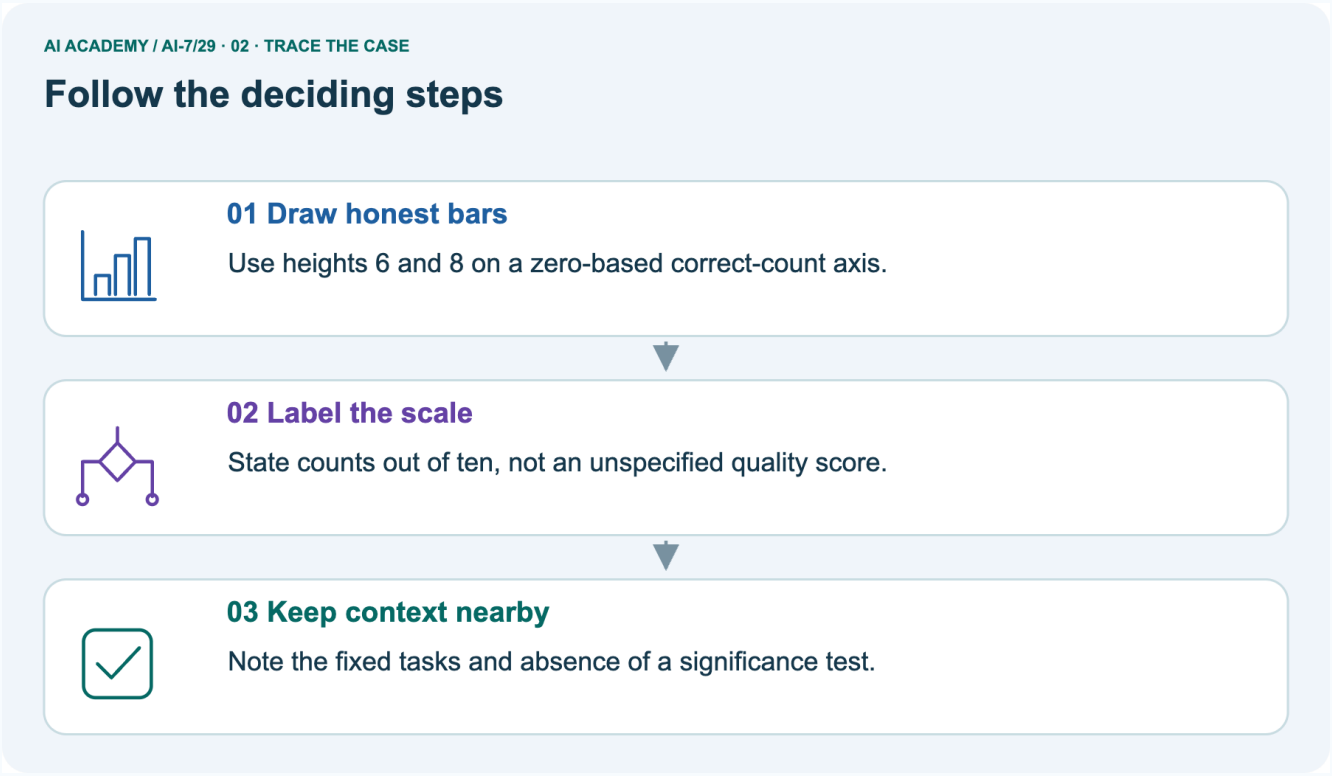
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases. Supplied facts: A has 6/10 correct; B has 8/10.

The count axis should start at zero and identify the ten-case denominator.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



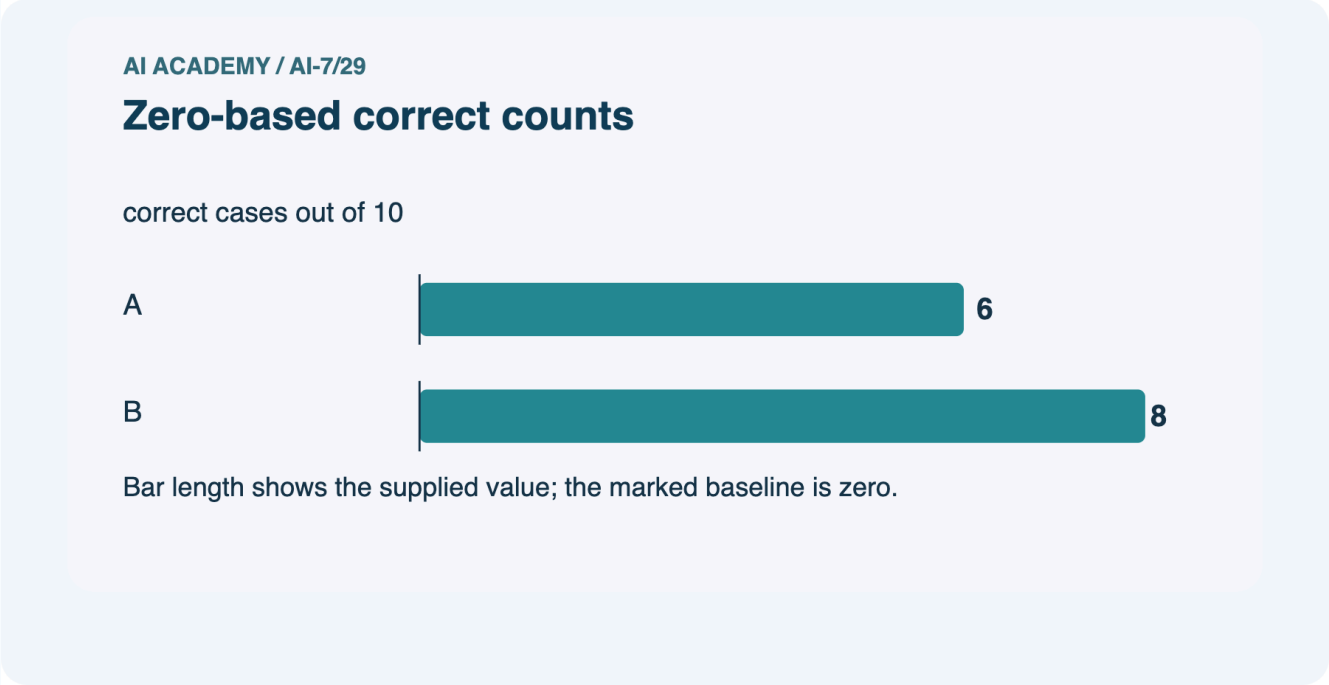
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw honest bars: Use heights 6 and 8 on a zero-based correct-count axis. → Label the scale: State counts out of ten, not an unspecified quality score. → Keep context nearby: Note the fixed tasks and absence of a significance test.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Zero-based correct counts



Follow the labelled picture with the matching explanation. Open the vector diagram

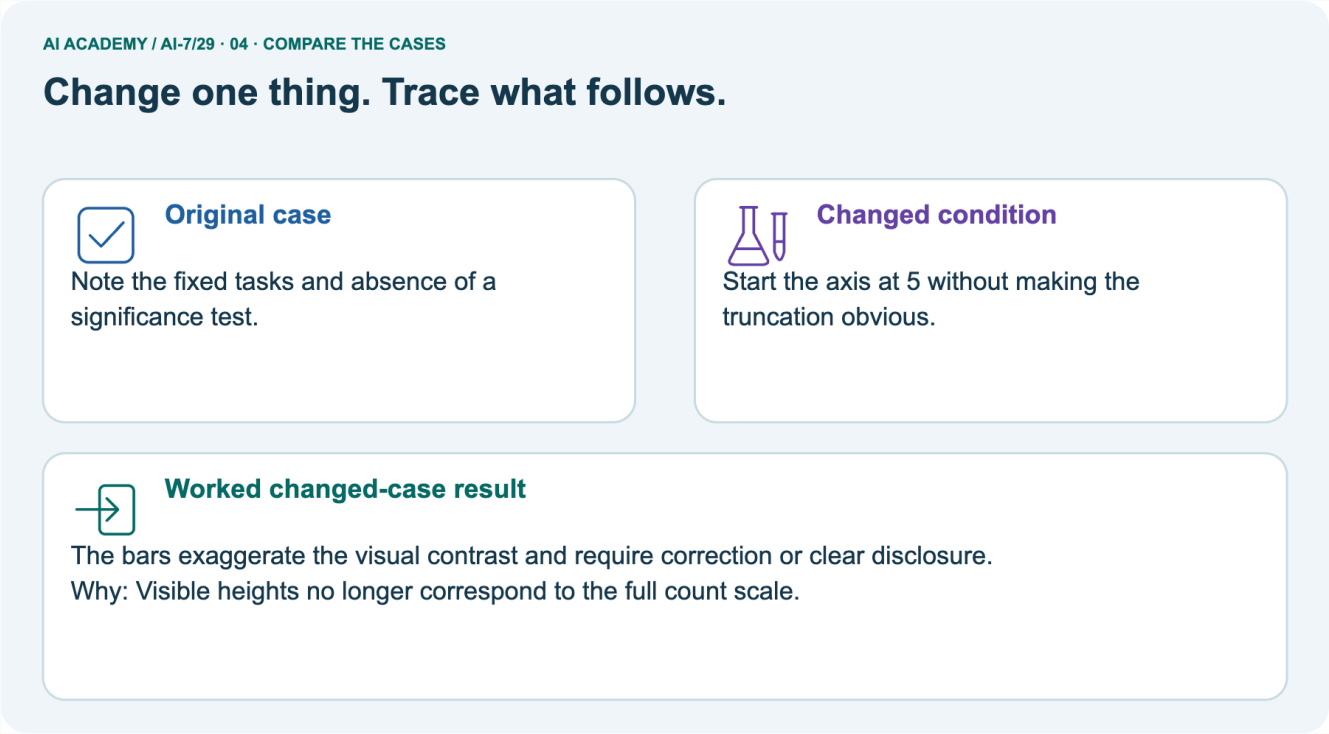
Zero-based correct counts. Units: correct cases out of 10. A: 6. B: 8.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Note the fixed tasks and absence of a significance test. Changed condition: Start the axis at 5 without making the truncation obvious.

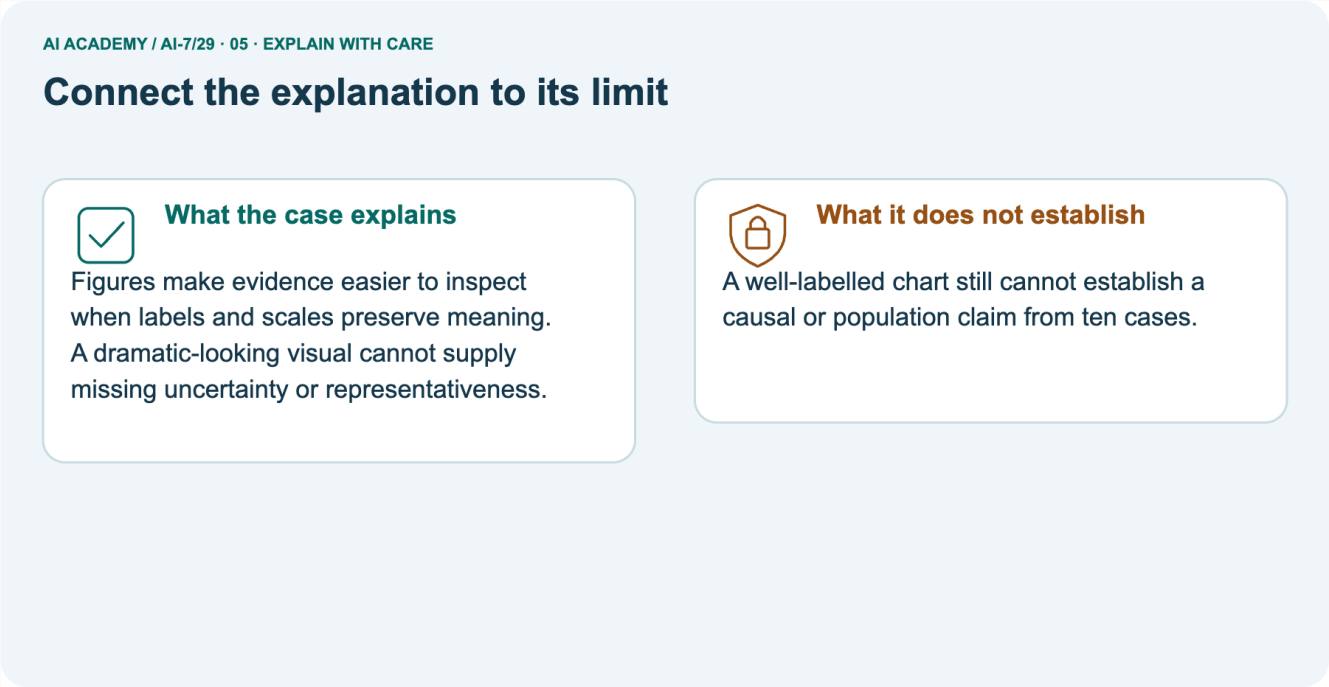
Expected result: The bars exaggerate the visual contrast and require correction or clear disclosure. Reason: Visible heights no longer correspond to the full count scale.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Visible heights no longer correspond to the full count scale.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

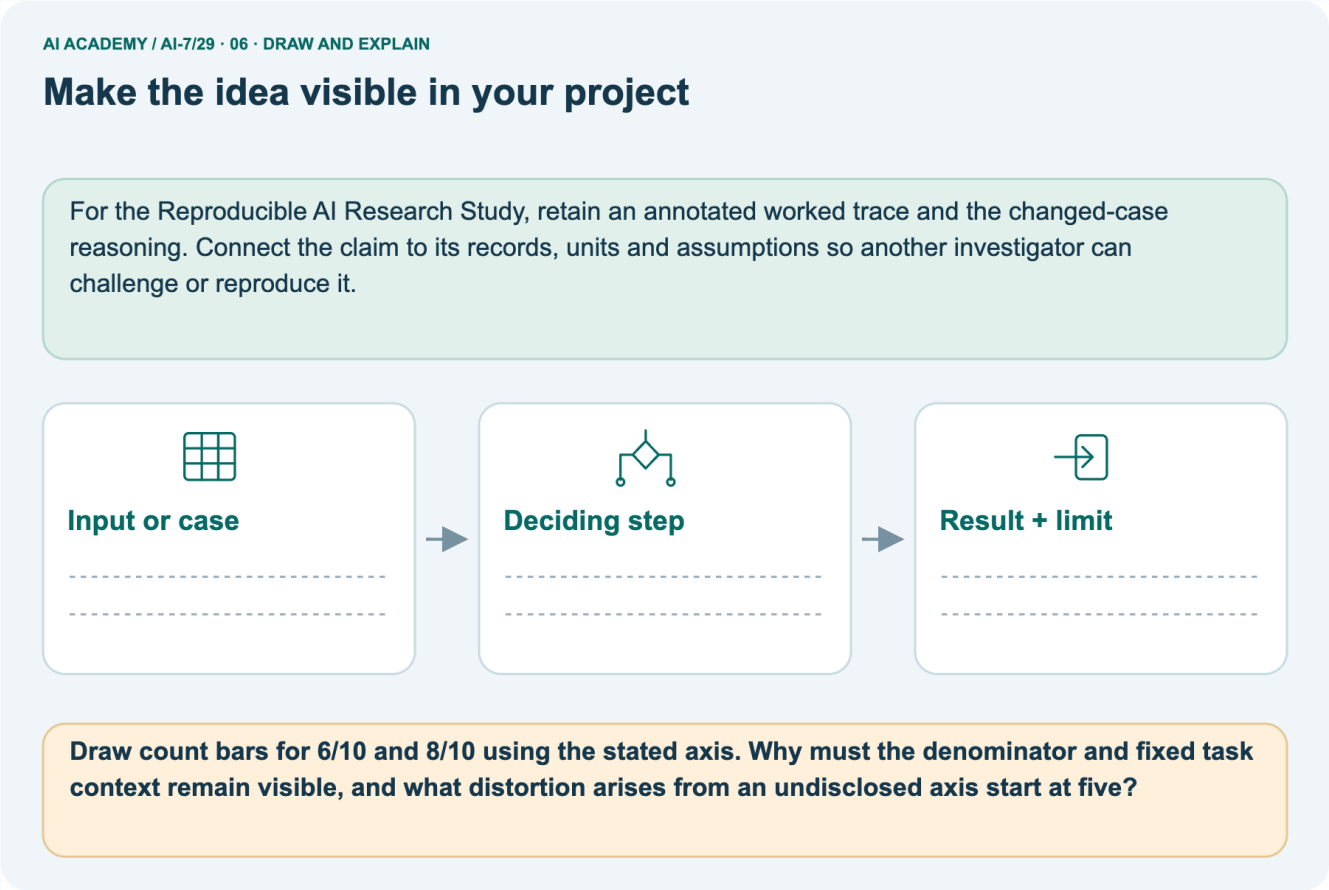
Limit: A well-labelled chart still cannot establish a causal or population claim from ten cases.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A well-labelled chart still cannot establish a causal or population claim from ten cases.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Use heights 6 and 8 on a zero-based correct-count axis. State counts out of ten, not an unspecified quality score. Note the fixed tasks and absence of a significance test.

Changed case: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Figures that preserve evidence: the case

02. Follow the deciding steps

03. Zero-based correct counts

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.